SUSTAINABLE AVIATION FUEL · KALIMANTAN

From biomass.
To boarding gates.

SAF Command makes infrastructure decisions inspectable.

80
supplied locations

6,800 kt
available biomass

4 airports
PNK · PKN · BPN · TRK

Joint site selection, facility sizing and material flows.

One connected planning decision.

Dispersed supply

Feedstock quantity and geography affect transport, facility placement and capacity needs.

Coupled investments

Depots and refineries must serve airport demand while respecting supply and mass conversion.

Transparent scope

13 screened depot candidates
8 screened refinery candidates
2 facility sizes per site
3 material-flow stages

Candidate coordinates are observed reference locations, not permitted construction land.

A real mixed-integer model.



Objective

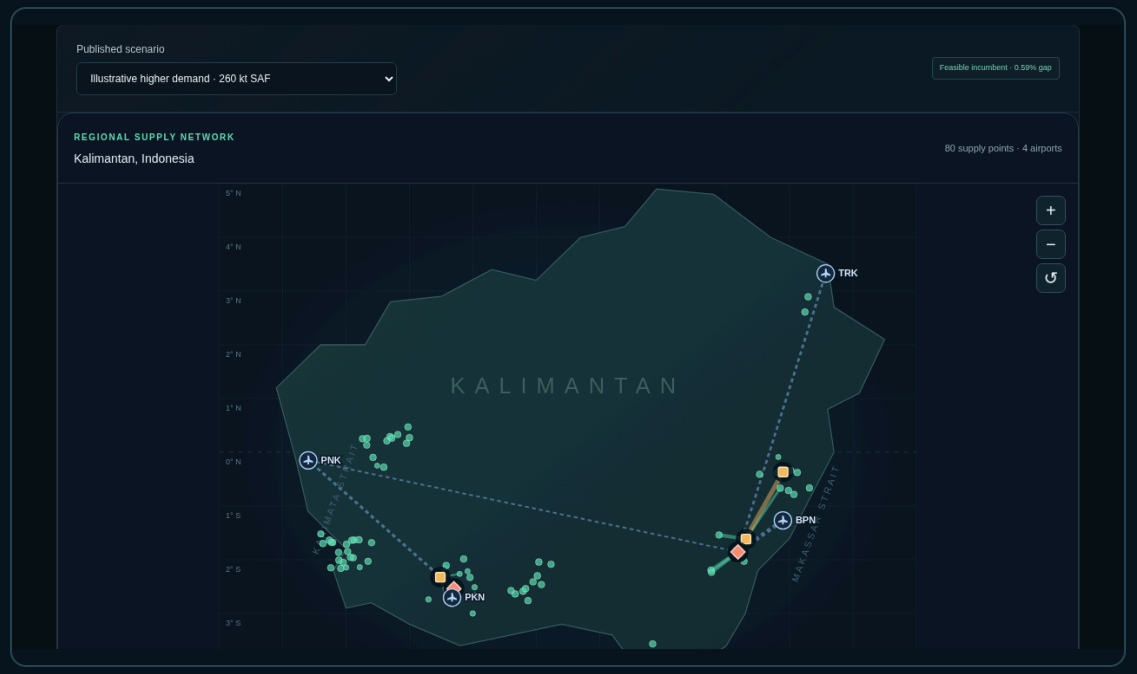
Facility CapEx + discounted recurring transport and configured processing costs.

Constraints

Supply, capacities, exclusive sizes, mass conservation, conversion yield and exact airport deliveries.

SciPy / HiGHS chooses facilities and quantities. Status, gap and assumptions remain visible.

Inspect the calculated network.



Actual static-export interface; published solver data.

Illustrative baseline

210 kt SAF demand
20% assumed mass yield
1,050 kt biomass input

US$800.99M
Modelled one-period objective
1.48% actual relative MIP gap

This example is not a verified investment budget or an exact global optimum. Costs can vary with solver termination within tolerance.

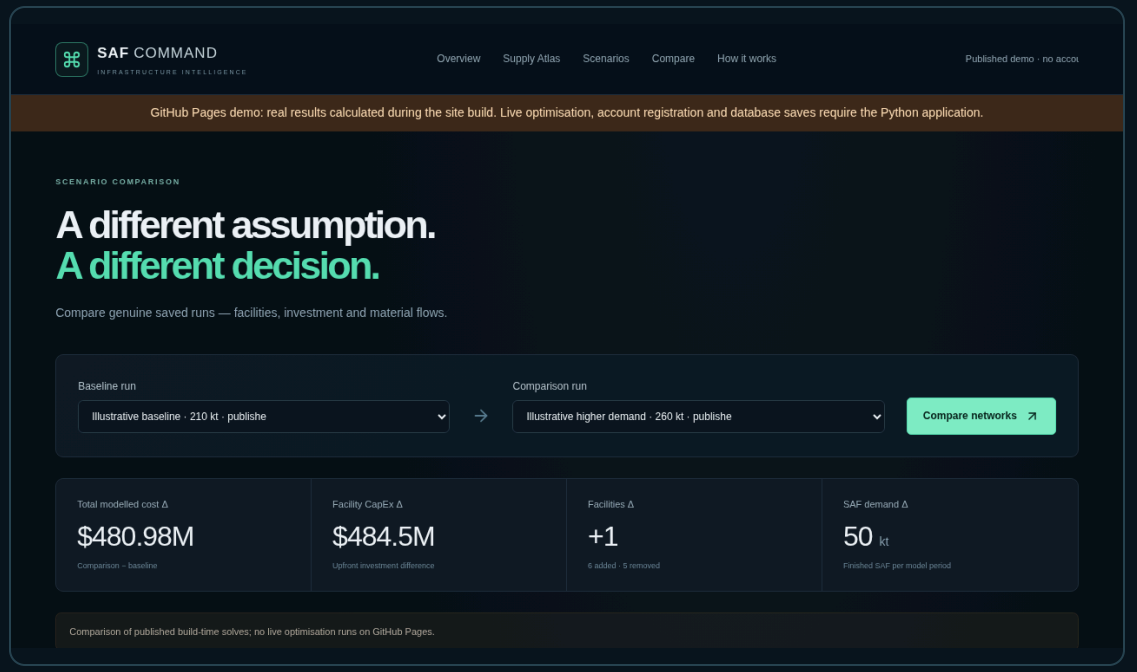
Change demand.
Inspect the consequences.

210 → 260 kt SAF

Compare actual facility choices and modelled costs. Trace quantities and export the inputs and result records.

Available biomass is a capacity limit; it does not need to be fully consumed.

The free Pages demonstration compares precomputed scenarios. Editable reruns are in the full Python application.



Ready to inspect.
Ready to reproduce.

Public demo

Atlas · published networks · comparison · explanations · downloads

Full application

Live Python solves · editable scenarios · accounts · PostgreSQL history

https://w0mbler.github.io/SAF-Anthropic-Hackathon/

Validation

70 backend tests
11 frontend tests
34 deployment-adapter tests
3 full-stack browser journeys
Separate static-only browser check
Hosted CI and Pages publication passed

These checks ran in their documented environments, not as a single combined suite.

Next: measured demand, verified routes, permitted sites, process economics and lifecycle evidence. Data licences remain separate from source-code MIT. No Claude/Anthropic integration claimed.